AI-Powered Full Stack Developer Program

CLASSWORK BOOKLET

Week 2 · Day 1

How the Web Works + HTML Core

Every classwork task in order: D1-00 … then fast finishers FF1–FF6 (no solutions)

Momo House · TSI Canteen · Aarav · Sunita · Didi

Stage 1, AI as Tutor: AI may explain; you type every line.

Tech Skills Institute (TSI) Nepal · Kathmandu

D1-00 · Setup check

File: classwork/D1-00-setup-check/README.md

Time: 8 min · **Type:** 🔨 Build · **Work:** alone

Goal: Open the pack in VS Code, see your page live, and open F12.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

1. Open VS Code. Click **File → Open Folder…** and choose the **`W2D1`** folder. Click **Select Folder**. (Open the *folder*, not a single file. Then all the links and tools work.)
2. In the Explorer on the left, open `classwork/D1-00-setup-check/index.html`.
3. Right-click inside the code → **Open with Live Server**. Chrome opens the page.
4. In VS Code, change `Student` in the `<h1>` to **your own name**.
5. Press **Ctrl+S** to save. Look at Chrome: the page reloads by itself.
6. In Chrome, press **F12**. Click the **Elements** tab. Find your `<h1>` with your name in it.
7. Put VS Code and Chrome side by side (Win+← and Win+→). You will work like this all day.

Done looks like

- Chrome shows "Namaste! My name is **your name**".
- You saved once and the page changed **without** pressing F5 (Live Server).
- F12 → Elements shows your `<h1>`.

Stuck?

1. No **Open with Live Server** in the menu? Use the fallback: double-click `index.html` in Windows Explorer. It opens in Chrome. After each save, press **F5**. Install Live Server at the break (see `START-HERE.md`).
2. The page did not change? Check the dot on the VS Code tab. A white dot means "not saved". Press Ctrl+S.
3. Still stuck? Raise your hand. One install problem must not stop you. Use the fallback and keep going.

Fast?

Read `START-HERE.md` again and check that **HTMLHint** is installed. You need it from block B3.

D1-01 · See the letters arrive

File: classwork/D1-01-see-the-letters-arrive/README.md

Time: 8 min · **Type:** 🔍 Explore · **Work:** alone

Goal: See with your own eyes that a web page is many parcels sent by a server.

Open it

This exercise uses a terminal, Live Server and Chrome DevTools. No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Remember the postal system: your browser writes a letter (the **request**). The server mails back a parcel (the **response**). One page = many parcels.

1. **Ping (a letter there and back).** Open a terminal in VS Code: menu **Terminal → New Terminal** (shortcut: **Ctrl** + the key under Esc).
   - No Wi-Fi: type `ping localhost` and press Enter.
   - Wi-Fi: type `ping google.com` and press Enter.

Copy one reply line into `my-findings.md`.

2. **Count the parcels.** Open `index.html` with Live Server. Press **F12** → **Network** tab → press **F5**. Count the lines. Write the number and the names in `my-findings.md`.
   - Do **not** count a line called `ws`. That is Live Server listening for your saves.
3. **Read the letter's answer.** Right-click on the page → **View Page Source** (or Ctrl+U). Find the `<h1>`. Write its text and its line number in `my-findings.md`.
4. **Wi-Fi bonus:** open any real website (for example a news site), F12 → Network → F5. How many requests?

Done looks like

- `my-findings.md` has a ping reply line, the number of parcels (it should be **7**), and the `<h1>` text and line number.
- You can say in one sentence: "One page is many files; the browser asks for each one."

Stuck?

1. The Network tab is empty? Keep it open and press F5 again. It only records while it is open.
2. You count 8? One line is `ws` (Live Server). Don't count it. Without Live Server (double-click), the Network tab shows the same 7 files as `file://`.
3. Ping says "could not find host"? No Wi-Fi. Use `ping localhost` instead.

Fast?

Open `style.css` and change the `h1` colour from `#990011` to `#0B6E4F`. Save. Which parcel changed? Check the Network tab.

D1-02 · Emmet speed drill

File: classwork/D1-02-emmet-speed-drill/README.md

Time: 7 min · **Type:** 🔨 Build · **Work:** alone

Goal: Use Emmet (speed-dial for tags) and 4 shortcuts to build a page fast.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

`index.html` is empty. Make it look like `target.html`. Open `target.html` with Live Server in a second tab to see the goal.

Type each abbreviation, then press **Tab**. Then type the words inside.

#	Type this	Then press	You get
1	`!`	Tab	the full skeleton (change the `<title>` to `Emmet Speed Drill`)
2	`h1+p`	Tab	a heading and a paragraph next to it
3	`ul>li*5`	Tab	a list with 5 items
4	`nav>a*3`	Tab	3 links in a nav (type `#` inside each `href=""`)
5	`table>tr*3>td*2`	Tab	a table: 3 rows, 2 cells each
6	`.card>h2+p`	Tab	a `<div class="card">` with a heading and a paragraph
Bonus 7	`ul>li.item$*3`	Tab	`item1`, `item2`, `item3`
Bonus 8	`lorem20`	Tab	20 words of dummy text

Shortcuts to use on the way:

Keys	What it does
Alt+Shift+↓	copy the line down
Ctrl+/	comment / uncomment the line
Shift+Alt+F	tidy up (format) the whole file
Ctrl+D	select the next match of a word (edit both at once)
Linked editing	change `<ul>` to `<ol>`: the closing tag changes too

Done looks like

- Your page in Chrome looks the same as `target.html`.
- You used all 6 required abbreviations and at least 3 shortcuts.
- Shift+Alt+F made your file tidy (2-space indents).

Stuck?

1. Tab does nothing? Check the bottom-right of VS Code says **HTML**. If it says Plain Text, click it and choose HTML.
2. Type the abbreviation with **no spaces**. `ul > li` does not work. `ul>li*5` does.
3. Open `target.html` in VS Code and compare line by line.

Fast?

Try `ul>li{Momo $}*4` and press Tab. What do the `{}` do?

D1-03 · Skeleton from memory

File: classwork/D1-03-skeleton-from-memory/README.md

Time: 4 min · **Type:** 🔨 Build · **Work:** alone

Goal: Type the whole HTML skeleton from memory. No Emmet, no notes.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Close your notes. Don't use `!`. Type every line yourself in the empty `index.html`:

1. The DOCTYPE line
2. `<html>` with the language attribute
3. `<head>` with: meta charset, meta viewport, title
4. `<body>` with one `<h1>` and one `<p>`
5. Close everything (last box opened, first box closed)

Wrong is fine. Blank is not. Then open the page with Live Server and check it against your notes.

Done looks like

- The tab shows your title.
- The page shows your heading and paragraph.
- The Problems panel (Ctrl+Shift+M) shows 0 problems.

Stuck?

1. Start with the boxes: `<html>`, then `<head>` and `<body>` inside it. Fill them after.
2. Can't remember the meta lines? Write the ones you remember. Check them later.
3. After 4 minutes, open `notes/03-html-skeleton-and-anatomy.md` and correct your file. Mark what you forgot.

Fast?

Delete everything and type it again. Can you do it in under 90 seconds?

D1-04 · Broken Skeleton

File: classwork/D1-04-broken-skeleton/README.md

Time: 10 min · **Type:** ⚽ Bug-hunt · **Work:** alone

Goal: Repair the skeleton of a page. Use HTMLHint and F12 to see the silent problems.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Aarav wrote his first page last night. Some parts are broken. The browser is a polite guesser: it shows *something*, but not what Aarav wanted.

File: `index.html` (one file).

There are 6 bugs. Find at least 4.

> ⚠ **Fix this first:** if most of the page is giant and bold, repair the heading first. It makes the other problems hard to see.

For each bug: find it, fix it, save, and check the page again.

Hints (one per bug type)

- **First line**: what must every HTML file start with?
- **The `<html>` tag**: it should say which language the page is in.
- **The browser tab**: does it show the page name? Check the spelling of the tag in `<head>`.
- **Giant text**: a heading that never ends.
- **Nesting**: last box opened, first box closed.
- **Missing footer**: how does a comment end?

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Done looks like

- No squiggles in VS Code. The Problems panel (Ctrl+Shift+M) shows **0** problems for this file.
- The tab says **Aarav's Page**.
- Only "Hello, I am Aarav" is big.
- The line and "Made by Aarav, Week 2" show at the bottom.

Scoreboard

Write this in your notebook:

```
I found __ / 6. The reveal taught me: ________
```

Stuck?

1. Open the Problems panel (Ctrl+Shift+M). Click each problem. VS Code jumps to the line.
2. Press F12 → Elements. Compare what the browser built with your code.
3. Compare with the skeleton in your notes (`notes/03-html-skeleton-and-anatomy.md`) line by line.

Fast?

Try **D1-FF4 Evil page** in `classwork/fast-finishers/`.

D1-05 · Grandmother's note

File: classwork/D1-05-grandmothers-note/README.md

Time: 4 min · **Type:** Modify · **Work:** alone

Goal: Give plain text a tone of voice with strong, em, br and blockquote.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Hajur Aama (grandmother) wrote Aarav a note. Right now it is one flat block of text. Give it her tone of voice:

- ☐ `<strong>`: the part she says is **important** ("the best momo in all of Kathmandu")
- ☐ `<em>`: the words she would **stress** when speaking ("do not")
- ☐ `<br>`: a pause inside one thought (and between "With love," and her name)
- ☐ `<blockquote>`: grandfather's words. She is **quoting someone else**
- ☐ Bonus: split the long paragraph into 3–4 short `<p>`s, and add an `<hr>` before "With love"

Done looks like

- One bold phrase, one stressed (italic) phrase, an indented quote.
- "With love," and "Hajur Aama" are on two lines.

Stuck?

1. `<strong>` and `<em>` go **inside** a `<p>`: `<p>… <strong>the best momo</strong> …</p>`.
2. `<br>` is a void tag: no closing tag.
3. `<blockquote>` holds a `<p>` inside it, like a box inside a box.

Fast?

Go on to **D1-06 Canteen News**.

D1-06 · Canteen News

File: classwork/D1-06-canteen-news-bughunt/README.md

Time: 10 min · **Type:** ⚽ Bug-hunt · **Work:** alone

Goal: Repair heading, entity and closing-tag mistakes in a news page.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Didi writes the **TSI Canteen Weekly** every Sunday. This week the page looks strange. Help her.

File: `index.html` (one file).

There are 5 bugs. Find at least 3.

For each bug: find it, fix it, save, and check the page again.

Hints (one per bug type)

- **Headings**: how many `<h1>` should one page have?
- **Entities**: how do you *print* a `<` or `>` on the page? Look for one that is spelled wrong, too.
- **Indent that never stops**: check how the quote ends.
- **Italic that never stops**: check every opening inline tag has its closing tag.

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Done looks like

- One `<h1>` only.
- The tip shows `Wrap text in <p> tags.` on one line.
- The price shows `Rs. 150`.
- Only Didi's words are indented. Only "do not" is italic.
- Problems panel: 0 problems.

Scoreboard

Write this in your notebook:

```
I found __ / 5. The reveal taught me: _________
```

Stuck?

1. Read the page from top to bottom in Chrome. Where does it *start* to look wrong? The mistake is just above that point.
2. Open the Problems panel (Ctrl+Shift+M). Two of the bugs are listed there.
3. Check the entities table in `notes/04-text-tags-and-entities.md`.

Fast?

Try **D1-FF1 Entity art** in `classwork/fast-finishers/`.

D1-07 · Broken Lists

File: classwork/D1-07-broken-lists/README.md

Time: 7 min · **Type:** 🐞 Bug-hunt · **Work:** alone

Goal: Repair the structure of four lists.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Aarav is cooking momo for his friends. His shopping list, spice box, steps and kitchen words are all lists, and some of them are broken.

File: `index.html` (one file).

There are 4 bugs. Find at least 2.

For each bug: find it, fix it, save, and check the page again.

Hints (one per bug type)

- **Missing parent**: look for the missing indent (and the missing bullets).
- **Order matters**: shopping list or recipe steps? Which list type gives numbers?
- **Typo**: read every list item tag letter by letter.
- **Glossary**: `dl` holds `dt` (the word) and `dd` (the meaning). Nothing else.

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Done looks like

- Every list is indented the same way.
- The cooking steps are numbered 1–4.
- "Garlic" has a bullet.
- The kitchen words show the meaning indented under each word, with no bullets.

Scoreboard

Write this in your notebook:

```
I found __ / 4. The reveal taught me: _________
```

Stuck?

1. Press F12 → Elements. Open `<body>`. Is every `<li>` inside a `<ul>` or `<ol>`?

2. Shopping list = order does not matter. Recipe steps = order matters.
3. Compare with the `<dl>` example in `notes/05-lists.md`.

Fast?

Go on to **D1-08 Momo recipe**.

D1-08 · Momo recipe

File: classwork/D1-08-momo-recipe/README.md

Time: 5 min · **Type:** 🔨 Build · **Work:** alone

Goal: Choose the right list for the job: ul, ol, a nested list, and (bonus) dl.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

The recipe text is inside a comment in `index.html`. Build it as real HTML:

- ☐ `<h2>Ingredients</h2>` + a `<ul>` (shopping list: order does not matter)
- ☐ Inside the **Spices** `<li>`, a **nested** `<ul>` with 4 spices
- ☐ `<h2>Steps</h2>` + an `<ol>` (recipe steps: order matters)
- ☐ One `<img>` of `images/momo-plate.svg` with a good `alt`
- ☐ Bonus: a `<dl>` "Kitchen words" glossary with 2 words (`dt`) and their meanings (`dd`)
- ☐ Delete the comment when you are done

Done looks like

- Bullets for ingredients, with the 4 spices indented under "Spices".
- Numbers 1–4 for the steps.
- The momo plate picture shows.

Stuck?

1. Emmet helps: `ul>li*5` then Tab.
2. The nested list goes **inside** the Spices `<li>`, before its `</li>`.
3. Look at the nested list example in `notes/05-lists.md`.

Fast?

Try **D1-FF2 Table of contents** in `classwork/fast-finishers/`.

D1-09 · Broken Paths

File: classwork/D1-09-broken-paths/README.md

Time: 9 min · **Type:** Bug-hunt · **Work:** alone

Goal: Repair broken links and image paths across folders.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Momo House has a small website with a `pages/` folder and an `images/` folder. Some links go nowhere and some pictures are broken. Click **every** link and look at **every** picture.

Files:

```
D1-09-broken-paths/
├── index.html
├── images/    (logo-momo-house.svg, momo.svg, thamel.svg)
└── pages/     (about.html, menu.html)
```

There are 6 bugs. Find at least 4.

For each bug: find it, fix it, save, and check the page again.

Hints (one per bug type)

- **Folder names**: is the folder in the path spelled exactly like the real folder?
- **Same house**: a link to a page in a sub-folder needs the folder name.
- **Postal address**: a link to another website needs the full address, starting with `https://`.
- **Go downstairs**: from inside `pages/`, use `../` to reach `index.html` and `images/`.
- **Only on my laptop**: a path with `C:\` works on one computer only.

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Done looks like

- All 3 pictures show (logo on Home, momo on Home and Menu, Thamel on About).
- Home → About → Home → Menu → Home works with no 404.
- The Wikipedia link opens wikipedia.org (Wi-Fi needed).

Scoreboard

Write this in your notebook:

```
I found __ / 6. The reveal taught me: _________
```

Stuck?

1. Hover over a link in Chrome. Look at the bottom-left corner: it shows where the link goes.
2. Open the VS Code Explorer. Follow the path with your finger: start at the file that has the link.
3. From `pages/about.html`, `../` means "go out of `pages/` first".

Fast?

Go on to **D1-10 Kathmandu gallery**.

D1-10 · Kathmandu gallery

File: classwork/D1-10-kathmandu-gallery/README.md

Time: 7 min · **Type:** 🔨 Build · **Work:** alone

Goal: Build a small gallery with figures, an external link and a "back to top" link.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

The pictures are ready in `images/`. Add:

- ☐ 2 `<figure>`s. Each has an `<img>` (with `src`, a good `alt` and `width="400"`) and a `<figcaption>`
  - `images/swayambhu.svg`
  - `images/patan-durbar.svg`
- ☐ 1 link to another website that opens in a **new tab**: `target="_blank"` (for example Kathmandu on Wikipedia)
- ☐ 1 "Back to top" link at the bottom: `<a href="#top">Back to top</a>`

Show & tell: after this, 2–3 pages go on the projector. Make yours worth showing!

Done looks like

- Two pictures with captions under them.
- The Wikipedia link opens a new tab.
- "Back to top" jumps to the top (make the window small to test it).
- HTMLHint: 0 problems. You will audit this folder again in **D1-16**.

Stuck?

1. `alt` = what you would say about the picture over the phone.
2. `<figure>` is the box; `<img>` and `<figcaption>` go inside it.
3. `#top` is special: every page has a top, so you don't need an `id` for it.

Fast?

Try **D1-FF6 Gallery detail page** (a sub-page with `../`) in `classwork/fast-finishers/`.

D1-11 · Broken Timetable

File: classwork/D1-11-broken-timetable/README.md

Time: 8 min · **Type:** Bug-hunt · **Work:** alone

Goal: Repair a table: header cells, cell counts, colspan and rowspan.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

This is the TSI Week 2 timetable. It has **6 columns**: Time + Sun, Mon, Tue, Wed, Thu. Every row must fill 6 columns. A `colspan` or `rowspan` cell counts for more than one.

File: `index.html` (one file). The `<style>` in `<head>` only draws lines around the cells. Leave it.

There are 5 bugs. Find at least 3.

For each bug: find it, fix it, save, and check the page again.

Hints (one per bug type)

- **Header row**: which tag makes a header cell?
- **Count the cells**: count every row. Each row must fill 6 columns.
- **Spelling**: check the attribute that joins cells across a row.
- **rowspan**: a cell that goes down 2 rows takes a place in the next row too.
- **Text outside cells**: all text in a table must be inside `<td>` or `<th>`.

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Done looks like

- The header row is bold.
- Every row lines up: no gaps, nothing sticks out on the right.
- Lunch and Tea break each fill the five day columns.
- Nothing shows above the table except the caption.

Scoreboard

Write this in your notebook:

```
I found __ / 5. The reveal taught me: ________
```

Stuck?

1. Put your finger on the screen and count the cells in each row.
2. Draw the table on paper: 6 columns. Mark which cells the `rowspan` and `colspan` take.
3. F12 → Elements: click a row (`<tr>`) to highlight it on the page.

Fast?

Try **D1-FF4 Evil page** in `classwork/fast-finishers/`.

D1-12 · Exam results table

File: classwork/D1-12-exam-results-table/README.md

Time: 6 min · **Type:** ✎ Modify · **Work:** alone

Goal: Turn a plain table into a well-built data table.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

The table has the data, but no structure. Add:

- ☐ a `<caption>` (the table's title): "TSI Week 1 quiz results (morning batch)"
- ☐ `<thead>` around the first row, and `<tbody>` around the 4 student rows
- ☐ header cells: change the first row's `<td>` to `<th scope="col">`
- ☐ a **Class total** row at the bottom: one cell with `colspan="3"` that says "Class total", and one cell with the sum (142)
- ☐ Bonus: put the Class total row in a `<tfoot>`

Done looks like

- A caption above the table, a bold header row, and a total row where "Class total" spans 3 columns.
- Every row fills exactly 4 columns.

Stuck?

1. The order inside `<table>` is: caption, thead, tbody, tfoot.
2. `colspan="3"` + 1 more cell = 4 columns. Count!
3. Look at the price-board example in `notes/08-tables.md`.

Fast?

Try **D1-FF3 Mystery page** in `classwork/fast-finishers/`.

D1-13 · Broken Signup

File: classwork/D1-13-broken-signup-form/README.md

Time: 10 min · **Type:** ⚽ Bug-hunt · **Work:** alone

Goal: Repair a signup form: labels, input types, radio names and values, and the submit button.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

This is the TSI course signup form. It uses `method="get"`, so when you submit it, your answers appear in the **address bar**. That is your tool for this hunt.

File: `index.html` (one file).

There are 6 bugs. Find at least 4.

> ⚠ **Fix this first:** the text box (textarea) is full of code. Repair the textarea first. It hides the last bug.

For each bug: find it, fix it, save, and check the page again.

Hints (one per bug type)

- **Labels**: click each label. Does the cursor jump into its box? `for` must match the input's `id`.
- **Input types**: is a phone number something you add and subtract? Try typing `+977 980` in the phone box.
- **Radio buttons**: can you pick only one batch? Radios in one group share one `name`.
- **What the kitchen receives**: submit and read the URL. Does it say which batch you chose?
- **Self-closing**: some tags cannot close themselves.
- **Submit**: a button must be inside the form to send it.

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Done looks like

- Clicking any label puts the cursor in its box.
- Phone has no arrows.
- Only one batch can be ticked.
- Fill the form, click **Sign up**, and the URL ends like:
  `?fullname=Sunita+Gurung&email=sunita%40example.com&phone=9841000000&batch=morning&message=I+love+HTML`

Scoreboard

Write this in your notebook:

```
I found __ / 6. The reveal taught me: ________
```

Stuck?

1. Submit and read the URL after `?`. Each `name=value` pair is one box.
2. Problems panel (Ctrl+Shift+M) will not help much here. Click, type and submit instead.
3. Compare each input with the examples in `notes/09-forms.md`.

Fast?

Try **D1-FF5 Notice Board** in `classwork/fast-finishers/`.

D1-14 · Extend the order slip

File: classwork/D1-14-extend-order-slip/README.md

Time: 7 min · **Type:** ✎ Modify · **Work:** alone

Goal: Add new kinds of inputs to a working form, and test each one in the URL.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

The TSI Canteen order slip works. Make it better.

Required (3):

- ☐ a **Phone** field with `type="tel"` and its `<label>`
- ☐ a **batch** radio group: Morning / Evening. Both radios share **one** `name` and have their own `value`
- ☐ a `<fieldset>` with a `<legend>Your batch</legend>` around the two radios

Bonus:

- ☐ a pickup date: `type="date"`
- ☐ two checkboxes: "Extra achar" and "Extra soup"
- ☐ `type="color"`: "Favourite colour for your TSI T-shirt"

Every new input needs a `<label>` with `for` = the input's `id`. Then fill the form, press **Send order**, and read the URL.

Done looks like

- Clicking each label puts the cursor in (or ticks) its input.
- Only one batch can be chosen.
- The URL shows your new fields, for example `…&phone=9841000000&batch=evening&….`

Stuck?

1. Copy an existing `<p>` block (Alt+Shift+↓) and change it. Faster and fewer typos.
2. Radios: same `name`, different `id` and `value`.
3. Nothing in the URL for a field? It has no `name`.

Fast?

Try **D1-FF5 Notice Board** in `classwork/fast-finishers/`.

D1-15 · Momo House repair (relay)

File: classwork/D1-15-momo-house-repair/README.md

Time: 17 min · **Type:** ⚽ Bug-hunt · **Work:** in pairs (relay)

Goal: Use everything from today to repair a whole 4-page website.

> Didi's Momo House website broke the night before opening. You are the repair crew.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

```
D1-15-momo-house-repair/
├── index.html    3 bugs
├── menu.html     3 bugs
├── order.html    2 bugs
├── about.html    2 bugs
└── images/       (logo-momo-house.svg, momo.svg, momo-plate.svg, chef-didi.svg)
```

There are 10 bugs. Find at least 8.

> ⚠ **Fix this first:** on `order.html`, the notes box is full of code. Repair that box first. It hides the rest of the page.

Relay rules

1. Work in pairs. One person types (the **driver**). One person reads and points (the **navigator**).
2. **Swap seats after every bug.**
3. Score: **1 point** per bug repaired. **+1 bonus** if you can name its visibility tag (V, D(HH), D(F12), R …).
4. Scores go on the whiteboard at **3:38** and **3:45**.
5. **Finished early?** Become a **helper** for another pair. Helpers may point. Helpers may **not** type. Then go to the fast finishers.

Hints (one per bug type)

- **Links**: click every link on every page. Check spelling and file endings.
- **Pictures**: every `<img>` needs `alt`. Is every path right from where the page is?
- **Closing tags**: if a heading looks like a link, a link never ended.
- **Tables**: count the columns. A `colspan` must fill the rest of the row exactly.
- **Lists**: every `<li>` needs a parent list.
- **Headings**: one level at a time (h1 → h2, never h1 → h4).
- **Forms**: click every label. Some tags cannot close themselves.
- **Other websites**: a full address starts with `https://`.

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint

Tag	Meaning	How you see it
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Scoreboard

Copy this on paper or the whiteboard:

Pair	index (3)	menu (3)	order (2)	about (2)	Bonus tags	Total

And in your notebook:

```
I found __ / 10. The reveal taught me: _________
```

Done looks like

- Home → Menu → Order → About → Home: every link works, every page.
- 4 pictures show: the logo on every page, momo on Home, momo plate on Menu, Didi on About.
- Only the heading "Welcome to Momo House" is big; it is **not** blue.
- The price table is a clean rectangle. Today's specials are indented.
- Click "Phone" on the order page: the cursor jumps into the box. Send an order and read the URL.
- The Problems panel (Ctrl+Shift+M) shows 0 problems for all 4 files.

Stuck?

1. Open the Problems panel (Ctrl+Shift+M). It finds 1–2 bugs for you.
2. Go page by page. The bug count per page is in the box above.
3. Ask a helper pair, or check today's bug-hunts: every bug here appeared once before (D1-04 to D1-13).

Fast?

Be a helper first. Then try **D1-FF6 Gallery detail page** in `classwork/fast-finishers/`.

D1-16 · Validate & self-audit

File: classwork/D1-16-validate-self-audit/README.md

Time: 4 min · **Type:** Modify · **Work:** alone

Goal: Check your own page with the HTML spell-checker, and repair every error.

Open it

Open your own **`classwork/D1-10-kathmandu-gallery/`** folder (the gallery you built). There are no new files here.

Your task

1. Open `D1-10-kathmandu-gallery/index.html` in VS Code.
2. Open the Problems panel: **Ctrl+Shift+M**. Repair every HTMLHint problem. Save after each one.
3. **Wi-Fi bonus:** go to **validator.w3.org → Validate by Direct Input**. Paste your whole file. Click **Check**. Repair every **Error** (you can leave Warnings).
4. Tick the audit checklist:

✔	Check	How
☐	The skeleton is complete: DOCTYPE, `html`, `head`, `body`	look at lines 1–10
☐	`<html lang="en">`	line 2
☐	meta charset + meta viewport + a real `<title>`	the tab shows your title
☐	Exactly **one** `<h1>`	Ctrl+F → `<h1`
☐	Headings go one level at a time	read them in order
☐	**Every** `<img>` has a useful `alt`	Ctrl+F → `<img`
☐	**Every** link works	click each one
☐	The external link opens a new tab	click it
☐	2-space indents, everything closed	Shift+Alt+F, then read
☐	Problems panel: **0**	Ctrl+Shift+M

Done looks like

- The Problems panel shows 0 problems for your gallery.
- All 10 boxes are ticked.

Stuck?

1. Click a problem in the Problems panel. VS Code jumps to the line.
2. Read the message slowly. "Tag must be paired" = something is not closed.
3. Compare with `notes/10-debugging-and-validating-html.md`.

Fast?

Audit another page from today (for example your **D1-14** order slip) with the same checklist.

Fast finishers · Day 1

File: classwork/fast-finishers/README.md

Finished the class exercise early? Pick one. They are in order, easiest first. You do **not** need to finish them all. No solutions in your pack: the instructor shows them after class.

Id	Title	Type	What you practise	About
D1-FF1	Entity art	Build	`< > & ©   → ♥`	10–15 min
D1-FF2	Table of contents	Build	nested lists, `#id` links, "back to top"	15 min
D1-FF3	Mystery page	Build	rebuild a page from a picture (`target.svg`)	15 min
D1-FF4	Evil page	8 bugs	bugs the browser hides (use F12 → Elements)	15 min
D1-FF5	Notice Board	12 bugs	one of every bug type from today	20–25 min
D1-FF6	Gallery detail page	Build	a sub-page with `../` and a `mailto:` link	15 min

When to try which (the instructor will also tell you):

- After **D1-06** → D1-FF1
- After **D1-08** → D1-FF2
- After **D1-10** → D1-FF3 or D1-FF6
- After **D1-11** → D1-FF4
- After **D1-13** → D1-FF5
- After **D1-15** (and helping another pair) → D1-FF6

Each folder has a `README.md`. Open the folder's `index.html` with Live Server.

D1-FF1 · Entity art

File: classwork/fast-finishers/D1-FF1-entity-art/README.md

Time: 10–15 min · **Type:** 🔨 Build · **Work:** alone

Goal: Print special characters with entities (escape codes).

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Build two things in `index.html`:

1. **A price board** for Didi: three lines like `Steam momo → Rs. 150`.
   - Use `→` for the arrow and ` ` between `Rs.` and the number (so they never split onto two lines).
   - Write "Momo & milk tea combo" with `&`, and add a `♥`.
2. **A tiny HTML lesson:** a paragraph that shows the tags **as text**: `A paragraph starts with <p> and ends with </p>.` Use `<` and `>`.
3. A footer line with `©` and the year.

Entity	Shows
`< >`	< >
`&`	&
`©`	©
` `	a space that never breaks
`→`	→
`♥`	♥

Done looks like

- The page shows arrows, a heart, a © sign and the text `<p>` — and no squiggles in VS Code.

Stuck?

1. Every entity starts with `&` and ends with `;`.
2. If you see `→` on the page as text, check the spelling and the `;`.
3. Challenge: how do you print the text `&` itself? (Answer: `&`)

Fast?

Try **D1-FF2 Table of contents**.

D1-FF2 · Table of contents

File: classwork/fast-finishers/D1-FF2-table-of-contents/README.md

Time: 15 min · **Type:** 🔨 Build · **Work:** alone

Goal: Build a table of contents that jumps to each part of a long page.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

The TSI student handbook is long. Help readers jump around.

1. Give every `<h2>` and `<h3>` an `id` (short, lowercase, with `-` for spaces, e.g. `id="morning-batch"`).
2. Under the intro, add `<h2>Contents</h2>` and a **nested list** of links:
   - each `<h2>` is an item: `<li><a href="#classes">Classes</a></li>`
   - its `<h3>`s are a sub-list **inside** that `<li>`
3. At the end of each section add `<a href="#top">Back to top</a>`.

Done looks like

- Clicking "Borrowing books" in the contents jumps down to that heading.
- "Back to top" jumps up again. The address bar shows `#borrowing`, then `#top`.

Stuck?

1. `href="#classes"` jumps to the element with `id="classes"`. No `#` inside the id itself.
2. The nested `<ul>` goes inside the `<li>`, before `</li>`.
3. Make the browser window short to see the jumps.

Fast?

Try **D1-FF3 Mystery page**.

D1-FF3 · Mystery page

File: classwork/fast-finishers/D1-FF3-mystery-page/README.md

Time: 15 min · **Type:** 🔨 Build · **Work:** alone

Goal: Rebuild a page from a picture: choose the right tag for each part.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

Open `target.svg` in Chrome (drag it into a tab, or right-click → Open with Live Server). It is a **picture** of a web page. Rebuild it in `index.html`:

- ☐ the `<title>` and the `<h1>`
- ☐ Didi's picture: `images/chef-didi.svg`, `width="150"`, with a good `alt`
- ☐ two lists: which one is a `<ul>` and which is an `<ol>`?
- ☐ the price table: Item | Price, with a header row
- ☐ the link "Visit Momo House": link it to `../../D1-15-momo-house-repair/index.html`

The table lines: copy the `<style>` block from `classwork/D1-11-broken-timetable/index.html` into your `<head>`.

Done looks like

- Put your page and `target.svg` side by side. Same text, same order, same kinds of tags.

Stuck?

1. Look at the shapes: bullets = `ul`, numbers = `ol`, a grid = `table`.
2. Bold, centred cells in a table = `th`.
3. Build top to bottom. Save and compare after each part.

Fast?

Try **D1-FF4 Evil page**.

D1-FF4 · Evil page

File: classwork/fast-finishers/D1-FF4-evil-page-bughunt/README.md

Time: 15 min · **Type:** ⚽ Bug-hunt · **Work:** alone

Goal: Find bugs that the browser hides from you.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

This page **looks almost fine**. That is what makes it evil. The browser is a polite guesser: it quietly repairs your mistakes in the Elements panel, so the page looks OK while the code is wrong.

File: `index.html`. Use **three tools**: your eyes, the Problems panel (Ctrl+Shift+M), and F12 → Elements.

There are 8 bugs. Find at least 6.

For each bug: find it, fix it, save, and check the page again.

Hints (one per bug type)

- **Pictures**: every `<img>` needs one attribute for people who can't see it.
- **Links**: a link cannot sit inside another link.
- **Boxes in the wrong box**: a `<p>` can't hold a list or a heading. A `<li>` needs a list parent.
- **ids**: each id is used once per page.
- **Forms**: click every word next to a box.
- **Entities**: read every `&…;` letter by letter.

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Done looks like

- F12 → Elements shows no empty `<p></p>` and no surprise tags.
- Problems panel: 0.
- Clicking "Plates" puts the cursor in the box.

Scoreboard

Write this in your notebook:

```
I found __ / 8. The reveal taught me: ________
```

Stuck?

1. Open F12 → Elements and look for empty `<p></p>` pairs. They mark where the browser repaired something.
2. Compare the Elements panel with your code, line by line.
3. Read the "browser repair table" in `notes/10-debugging-and-validating-html.md`.

Fast?

Try **D1-FF5 Notice Board**.

D1-FF5 · Notice Board

File: classwork/fast-finishers/D1-FF5-notice-board-bughunt/README.md

Time: 20–25 min · **Type:** Bug-hunt · **Work:** alone

Goal: Find the bugs that were cut from today's class hunts. Nothing is thrown away!

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

The TSI notice board page has one of every kind of bug from today: skeleton, text, lists, paths and tables.

Files: `index.html` and `images/notice-board.svg`.

There are 12 bugs. Find at least 10.

For each bug: find it, fix it, save, and check the page again.

Hints (one per bug type)

- **Skeleton**: check how `<head>` ends, and that nothing comes after `</html>`.
- **Headings**: one level at a time; headings are not for making text bold.
- **Spacing**: `<br>` is a pause, not a spacer. Is every void tag written correctly?
- **Lists**: a nested list lives inside an `<li>`; a list only holds `<li>`s; open and close with the same tag.
- **Paths**: file names have no spaces. A `#link` must match an `id` exactly.
- **Tables**: every cell lives in a row.

How bugs hide (the visibility legend)

Tag	Meaning	How you see it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code (Problems panel: Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser really built	F12 → Elements
D(URL)	Submit the form and read the address bar	the URL bar
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi)
H	Hidden on Windows; breaks on a real Linux server (capital letters in a file name)	the instructor explains
R	Rule check: no tool shows it. Read the code and check the rule	your brain + the notes

Done looks like

- The picture shows. The divider line shows. No big gap.
- "Events" jumps to the Events heading.
- Problems panel: 0. F12 → Elements matches your code.

Scoreboard

Write this in your notebook:

```
I found __ / 12. The reveal taught me: _________
```

Stuck?

1. Problems panel first (Ctrl+Shift+M). Then your eyes. Then F12 → Elements.

2. Each hint above is one or two bugs. Tick them off.
3. The bug types come from D1-04, D1-06, D1-07, D1-09 and D1-11. Look back at those reveals.

Fast?

Try **D1-FF6 Gallery detail page**.

D1-FF6 · Gallery detail page

File: classwork/fast-finishers/D1-FF6-gallery-detail-page/README.md

Time: 15 min · **Type:** 🔨 Build · **Work:** alone

Goal: Make a page inside a sub-folder and link it with `../`.

Open it

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file. It opens in Chrome. Press F5 after each save.

Your task

The gallery links to `places/swayambhu.html`, but that page does not exist yet. The `places/` folder is empty.

1. In VS Code, right-click the `places` folder → **New File** → `swayambhu.html`. Type `!` + Tab.
2. Add:
   - ☐ a link back to the gallery: `<a href="../index.html">Back to the gallery</a>` (go downstairs one floor)
   - ☐ an `<h1>` and 2 short paragraphs about Swayambhunath
   - ☐ the picture, using `../images/swayambhu.svg`, with `alt`
   - ☐ a `mailto:` link to the **TSI tour desk**: `<a href="mailto:tours@tsi.edu.np">TSI tour desk</a>`
3. Bonus: make `places/boudha.html` the same way and link it from the gallery caption.

Done looks like

- Gallery → Swayambhunath → Back to the gallery works both ways.
- The picture shows on the detail page.
- Clicking "TSI tour desk" opens your email app (or asks which app to use).

Stuck?

1. The detail page is one floor up (inside `places/`). Every path to something outside `places/` starts with `../`.
2. Broken picture? Hover over the `src` path in VS Code and Ctrl+click it. Does VS Code find the file?
3. Look at `notes/06-links-and-paths.md`: "go downstairs one floor".

Fast?

Make all three detail pages, and link each caption.